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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Library Book Borrowing, Return and Fine Calculation System

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25BCS150

Under the Guidance of

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DECLARATION

I hereby declare that the project entitled "**Library Book Borrowing, Return and Fine Calculation System"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **Mr.Aravind ,**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Mr.Nakul

CERTIFICATE

This is to certify that the project entitled **"Library Book Borrowing, Return and Fine Calculation System"** is a **bona fide work** carried out by **Mr.NAKUL [1Cs-B 2025**], **Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

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ABSTRACT

The Library Book Borrowing, Return and Fine Calculation System is a software application developed to automate and simplify the management of library activities. The main purpose of this system is to efficiently handle the process of issuing and returning books while maintaining accurate records of users and book details. In traditional library systems, manual record keeping leads to errors, time consumption, and difficulty in tracking due dates.

This system provides an easy and reliable solution by storing all necessary information digitally. It keeps track of issue and return dates and automatically calculates fines for late returns based on the number of delayed days. This reduces manual effort and ensures accuracy in calculations. The system is designed using Java programming language and provides a simple interface for users to interact with.

Overall, the system improves efficiency, reduces paperwork, and enhances the performance of library management. It is especially useful for schools and colleges to manage their library operations in an organized and effective manner.

1.INTRODUCTION

Library Book Borrowing, Return and Fine Calculation System is a software application designed to automate and simplify the management of library activities. In traditional libraries, all records are maintained manually, which leads to errors, confusion, and time consumption. This system helps in maintaining proper records of books, students, issue dates, return dates, and fines in an organized digital format. It improves efficiency and reduces paperwork.

The system allows users to borrow and return books easily without any complications. It keeps track of due dates and automatically calculates fines for late returns, which avoids manual mistakes. This system is especially useful for schools and colleges where managing large numbers of books and students can be difficult. It provides a user-friendly interface so that both librarians and students can use it without technical knowledge. Overall, the system ensures accuracy, saves time, and improves the overall performance of library management.

1.1 OBJECTIVE

The main objective of this system is to develop an automated solution for managing library operations efficiently. It aims to reduce manual work involved in maintaining records of issued and returned books. The system helps in tracking due dates and ensures that books are returned on time. It also calculates fines automatically when books are returned late. This improves accuracy and avoids human errors.

Another objective is to maintain a proper database of books and users. The system ensures that all information is stored securely and can be accessed easily when needed. It also aims to provide a simple and user-friendly interface for smooth operation. Overall, the objective is to improve efficiency, save time, and provide a reliable library management system.

1.2 OVERVIEW

The Library Book Borrowing, Return and Fine Calculation System provides a complete solution for managing library activities. It includes features like adding books, issuing books to users, returning books, and calculating fines. The system stores all relevant information such as book details, user details, issue date, and return date.

When a book is borrowed, the system records the issue date and sets a return deadline. If the book is returned after the due date, the system calculates the fine based on the number of delayed days. This eliminates the need for manual calculations and reduces errors. The system is designed in such a way that it is easy to use and understand, even for beginners.

1.3 FEATURES

Provides easy book borrowing and return process

Automatically calculates fine for late returns

Maintains records of books and users

Reduces manual work and paperwork

Improves accuracy and efficiency

Simple and user-friendly interface

Saves time for both librarian and users

2.PROGRAM ANALYSIS

The existing manual system of library management has several drawbacks. Maintaining records manually is time-consuming and increases the chances of errors. It becomes difficult to track borrowed books and calculate fines accurately. This leads to confusion and inefficiency.

To overcome these problems, the Library Book Borrowing, Return and Fine Calculation System is developed. This system automates all major library operations and provides accurate results. It helps in maintaining proper records and improves the efficiency of the system. The analysis shows that an automated system is necessary for better management of library activities.

2.1 PROBLEM DEFINITION

In manual systems, librarians have to write down all the details of issued and returned books in registers. This process is slow and prone to errors. Calculating fines manually can also lead to incorrect results. There is no proper tracking system, which makes it difficult to find information quickly.

The main problem is the lack of an efficient and accurate system to manage library operations. Therefore, there is a need for a computerized system that can handle all these tasks automatically and provide reliable results.

2.2 PROJECT OVERVIEW

This project focuses on developing a simple and effective system for managing library operations. It includes functionalities like issuing books, returning books, and calculating fines. The system stores all data in a structured format, making it easy to retrieve information when required.

The project is implemented using Java programming language. It provides a console-based interface where users can interact with the system. The system reduces workload and improves accuracy, making library management easier and faster.

2.3 SCOPE AND LIMITATIONS

The scope of this system includes managing book records, user details, and fine calculations efficiently. It can be used in schools and small college libraries. It helps in reducing manual errors and improving performance.

However, the system has some limitations. It is suitable only for small-scale use and may not handle large databases. It does not include advanced features like online access or barcode scanning. It also requires basic knowledge of computers to operate.

3.SYSTEM DESIGN

The system is designed using a modular approach where each function is divided into separate modules. This makes the system easy to understand, develop, and maintain. Each module handles a specific task and works together to complete the overall process.

3.1 MODULES

Book Management Module – manages book details

User Management Module – stores user information

Borrowing Module – handles book issuing

Return Module – manages book returning

Fine Calculation Module – calculates fine for delay

Data Storage Module – stores all records

Display Module – shows output to the user

4.METHODOLOGY

The methodology followed in this project includes analyzing the problem, designing the system, implementing the solution, and testing the output. A step-by-step approach is used to ensure proper development of the system.

4.1 PROBLEM ANALYSIS

The first step is to analyze the problems in the existing manual system. Issues like errors, time consumption, and difficulty in maintaining records are identified. Based on this analysis, requirements for the new system are defined.

4.2 SYSTEM DEVELOPMENT

The system development phase involves designing and coding the application using Java programming language. In this stage, all the modules such as book management, user management, borrowing, returning, and fine calculation are developed. Each module is implemented separately and later integrated to form the complete system. The system is

tested continuously to ensure proper working. Errors are identified and corrected during development. The focus is on creating a simple, efficient, and accurate system.

4.3 DATA COLLECTION

Data collection is an important step in developing this system. Information about books, users, issue dates, and return dates is collected. Sample data is used to test the system. The collected data helps in understanding how the system should work. It also helps in designing proper inputs and outputs. The data is organized in a structured manner for easy access and processing.

4.4 SYSTEM REQUIREMENTS

The system requires both hardware and software components to function properly.

Hardware Requirements:

Computer or Laptop

Minimum 4GB RAM

Keyboard and Monitor

Software Requirements:

Java Development Kit (JDK)

Any IDE like Eclipse or NetBeans

Operating System (Windows/Linux)

4.5 SYSTEM DESIGN APPROACH

The system follows a modular design approach. It is divided into different modules such as book management, user management, borrowing, returning, and fine calculation. Each module performs a specific function. The design ensures smooth flow of data between modules. It also improves readability and maintainability of the system. The approach makes the system easy to develop and test.

4.6 TESTING

Testing is done to ensure that the system works correctly. Different test cases are used to check all functionalities. The system is tested for book borrowing, returning, and fine calculation. Errors are identified and fixed during testing. Both normal and edge cases are tested. Testing ensures reliability and accuracy of the system.

4.7 IMPLEMENTATION PLAN

The implementation plan describes how the system is executed step by step. First, the program is written and compiled. Then, it is tested with sample inputs. After testing, the system is executed in real conditions. The output is verified for correctness. Any errors are corrected. Finally, the system is ready for use. The implementation plan ensures smooth execution of the project.

5.SOURCE CODE

```
import java.util.*;

class LibrarySystem {
    String bookName;
    String studentName;
    int borrowDay;
    int returnDay;
    int fine;

    Scanner sc = new Scanner(System.in);

    // Borrow Book
    void borrowBook() {
        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();
```

```
    System.out.print("Enter Book Name: ");
    bookName = sc.nextLine();

    System.out.print("Enter Borrow Day: ");
    borrowDay = sc.nextInt();

    System.out.println("Book Borrowed Successfully!\n");
}

// Return Book
void returnBook() {
    System.out.print("Enter Return Day: ");
    returnDay = sc.nextInt();

    calculateFine();
}

// Fine Calculation
void calculateFine() {
    int days = returnDay - borrowDay;

    if (days > 7) {
        fine = (days - 7) * 2; // ₹2 per day after 7 days
    } else {
        fine = 0;
    }

    displayDetails();
}
```

```
    // Display Details
    void displayDetails() {
        System.out.println("\n----- Library Details -----");
        System.out.println("Student Name: " + studentName);
        System.out.println("Book Name: " + bookName);
        System.out.println("Borrow Day: " + borrowDay);
        System.out.println("Return Day: " + returnDay);
        System.out.println("Fine: ₹" + fine);
    }
}

public class Main {
    public static void main(String[] args) {
        LibrarySystem obj = new LibrarySystem();

        obj.borrowBook();
        obj.returnBook();
    }
}
```

6.INPUT AND OUTPUT SCREENS

Figure 1: input screen

```
C:\Users\Student>java Main

Enter Student Name: Rahul
Enter Book Name: Computer Networks
Enter Borrow Day: 2
Enter Return Day: 11
```

Figure 2: output screen

```
----- Library Details -----
Student Name: Rahul
Book Name: Computer Networks
Borrow Day: 2
Return Day: 11
Fine: ₹8
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Library Book Borrowing, Return and Fine Calculation System was successfully developed and tested with different inputs. The system is able to perform all the main functions such as borrowing books, returning books, and calculating fines accurately. During testing, it was observed that the system provides correct results for both on-time and late returns. The discussion shows that the system reduces manual work and minimizes errors in maintaining records. It helps in keeping track of issued books and ensures proper management of return dates. The automatic fine calculation feature works efficiently and avoids mistakes that may occur in manual calculations. The system is simple to use and provides clear output, which makes it easy for users to understand. Overall, the system improves efficiency, saves time, and provides a reliable solution for library management. It meets all the objectives defined at the beginning of the project.7.2

7.2 Conclusion

The Library Book Borrowing, Return and Fine Calculation System is an effective solution for managing library operations. It simplifies the process of issuing and returning books and ensures proper record maintenance. The system reduces manual effort and increases accuracy in handling data.

The project successfully demonstrates how automation can improve efficiency in library management. It is user-friendly and easy to implement using Java. The system performs all required operations correctly and provides accurate results.

In conclusion, this project is useful for small and medium-sized libraries and can be further enhanced with advanced features like database integration, graphical user interface, and online access to make it more powerful and efficient.

8.REFERENCES

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